Week Two Assignment: First VMs - Louis Desgrée

For this assignment, I deployed two virtual machines Kali Linux and Metasploitable 2 that will be used in other assignments and projects. The purpose of this lab was to configure and verify isolated network communication between the two machines while also maintaining optional internet connectivity on Kali via a separate network adapter.

To begin, I imported both virtual machine images into VMware and verified that the deployments were successful by taking initial snapshots immediately after the installation. For Kali Linux, I ensured that two network adapters were configured. The first adapter was set to “Shared Network” (NAT mode) in order to provide optional internet access, and the second adapter was configured to operate in “Host Only” mode on a custom VMnet. In the network editor, I created a new host-only network (for instance, VMnet 5) with the subnet 10.10.10.0/24 and enabled DHCP (or manually set static IP addresses, if DHCP was not available).

For Metasploitable 2, I configured a single network adapter connected to the same custom host-only network. This ensured that the Metasploitable machine was isolated from the Internet, thereby reducing exposure while still permitting communication with Kali. Following the deployment of both VMs, I powered them on and accessed the terminal in each machine. Using the commands ip addr show and ifconfig, I verified that the Kali machine’s host-only interface received the IP address 10.10.10.5 and the Metasploitable machine was assigned the IP address 10.10.10.6 (or the corresponding addresses within the subnet).

To confirm proper connectivity, I executed ping tests from each VM. From the Kali terminal, I ran the command ping 10.10.10.6, and from the Metasploitable terminal, I ran ping 10.10.10.5. Both tests generated successful replies, thereby confirming that the two systems could communicate over the isolated network. I have included screenshots demonstrating each critical step: the virtual machine deployment process; the configuration of both network adapters (showing the settings in UTM’s network editor); the terminal outputs where the IP addresses were displayed; and the ping test results verifying that communication between the machines was successful.

In summary, the assignment required a systematic approach to setting up two distinct networks for Kali Linux, ensuring internet access is provided through one interface while isolating the lab network on another interface. Metasploitable was configured to use only the isolated network. By carefully following these steps and verifying the network configuration through snapshots and terminal output, I have successfully met all the assignment requirements. Attached below are the annotated screenshots of the deployment, network configuration, IP address display, and ping tests that document the complete process.

```
┌──(louis㉿kali)-[~]
└─$ sudo ip addr add 10.10.10.5/24 dev eth1
[sudo] password for louis:

┌──(louis㉿kali)-[~]
└─$ sudo ip link set eth1 up

┌──(louis㉿kali)-[~]
└─$ ip addr show dev eth1
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP grou
p default qlen 1000
    link/ether 0a:8a:5f:22:2f:36 brd ff:ff:ff:ff:ff:ff
    inet 192.168.128.3/24 brd 192.168.128.255 scope global dynamic noprefixroute
 eth1
       valid_lft 3544sec preferred_lft 3544sec
    inet 10.10.10.5/24 scope global eth1
       valid_lft forever preferred_lft forever
    inet6 fe80::88a:5fff:fe22:2f36/64 scope link noprefixroute
       valid_lft forever preferred_lft forever

┌──(louis㉿kali)-[~]
└─$ sudo ifconfig eth0 10.10.10.6 netmask 255.255.255.0 up

┌──(louis㉿kali)-[~]
└─$ sudo ip link set eth0 up
```

```
┌──(louis㉿kali)-[~]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.10.10.6  netmask 255.255.255.0  broadcast 10.10.10.255
        ether be:a5:ec:52:bc:41  txqueuelen 1000  (Ethernet)
        RX packets 56  bytes 25800 (25.1 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.128.3  netmask 255.255.255.0  broadcast 192.168.128.255
        inet6 fe80::88a:5fff:fe22:2f36  prefixlen 64  scopeid 0x20<link>
        ether 0a:8a:5f:22:2f:36  txqueuelen 1000  (Ethernet)
        RX packets 1  bytes 342 (342.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 24  bytes 3006 (2.9 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 24  bytes 1440 (1.4 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 24  bytes 1440 (1.4 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

```
┌──(louis㉿kali)-[~]
└─$ ping  10.10.10.6
PING 10.10.10.6 (10.10.10.6) 56(84) bytes of data.
64 bytes from 10.10.10.6: icmp_seq=1 ttl=64 time=0.085 ms
64 bytes from 10.10.10.6: icmp_seq=2 ttl=64 time=0.056 ms
64 bytes from 10.10.10.6: icmp_seq=3 ttl=64 time=0.066 ms
64 bytes from 10.10.10.6: icmp_seq=4 ttl=64 time=0.137 ms
^C
--- 10.10.10.6 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3078ms
rtt min/avg/max/mdev = 0.056/0.086/0.137/0.031 ms

┌──(louis㉿kali)-[~]
└─$ ping  10.10.10.5
PING 10.10.10.5 (10.10.10.5) 56(84) bytes of data.
64 bytes from 10.10.10.5: icmp_seq=1 ttl=64 time=0.051 ms
64 bytes from 10.10.10.5: icmp_seq=2 ttl=64 time=0.057 ms
64 bytes from 10.10.10.5: icmp_seq=3 ttl=64 time=0.051 ms
64 bytes from 10.10.10.5: icmp_seq=4 ttl=64 time=0.057 ms
^C
--- 10.10.10.5 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3083ms
rtt min/avg/max/mdev = 0.051/0.054/0.057/0.003 ms
```

```
       =[ metasploit v6.4.44-dev                          ]
+ -- --=[ 2487 exploits - 1281 auxiliary - 431 post       ]
+ -- --=[ 1466 payloads - 49 encoders - 13 nops           ]
+ -- --=[ 9 evasion                                       ]

Metasploit Documentation: https://docs.metasploit.com/

msf6 > ifconfig
[*] exec: ifconfig

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.10.10.6  netmask 255.255.255.0  broadcast 10.10.10.255
        ether be:a5:ec:52:bc:41  txqueuelen 1000  (Ethernet)
        RX packets 58  bytes 26084 (25.4 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.128.3  netmask 255.255.255.0  broadcast 192.168.128.255
        inet6 fe80::88a:5fff:fe22:2f36  prefixlen 64  scopeid 0x20<link>
        ether 0a:8a:5f:22:2f:36  txqueuelen 1000  (Ethernet)
        RX packets 1  bytes 342 (342.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 25  bytes 3068 (2.9 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

```
$ /usr/share/kali-menu/helper-scripts/metasploit-framework.sh
[sudo] password for louis:
[i] Database already started
[i] The database appears to be already configured, skipping initialization
Metasploit tip: Tired of setting RHOSTS for modules? Try globally setting it
with setg RHOSTS x.x.x.x
```